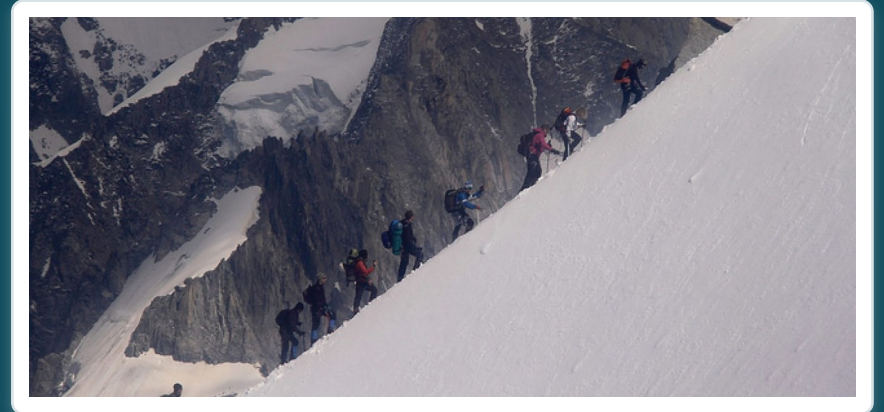


CHAPTER 22

The Respiratory System

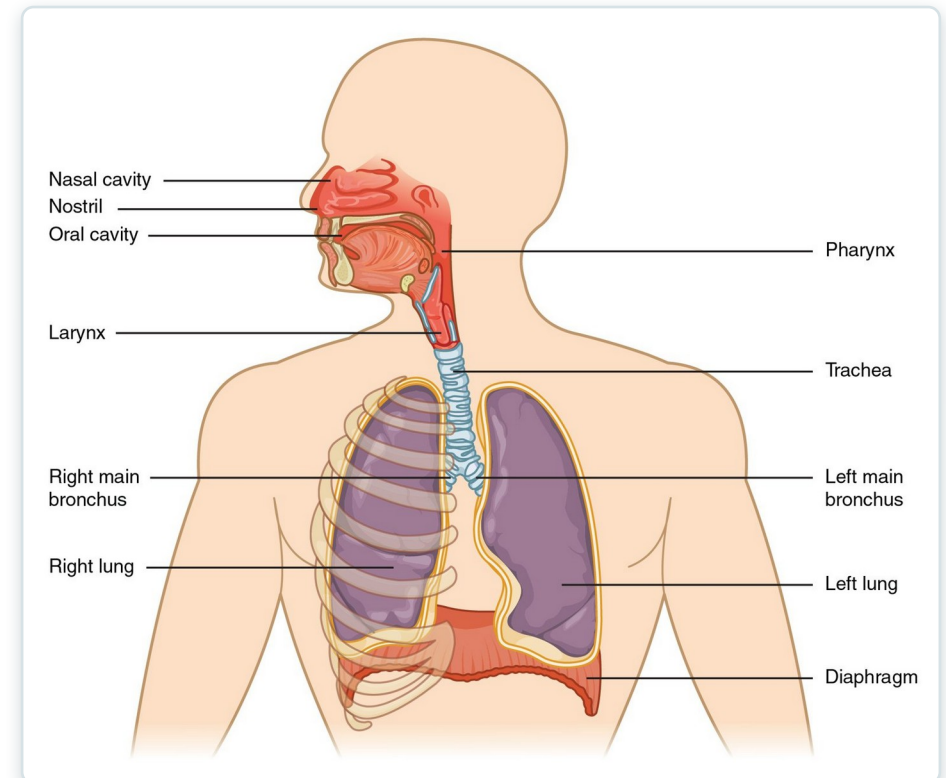
The airways, the lungs, breathing, and gas exchange.



Learning Objectives

After studying this chapter, students will be able to:

- 1 List the structures of the respiratory system
- 2 List the major functions of the respiratory system
- 3 Outline the forces that allow for air movement into and out of the lungs
- 4 Outline the process of gas exchange
- 5 Summarize the process of oxygen and carbon dioxide transport within the respiratory system
- 6 Create a flow chart illustrating how respiration is controlled



The organs of the respiratory system.



22.1

SECTION

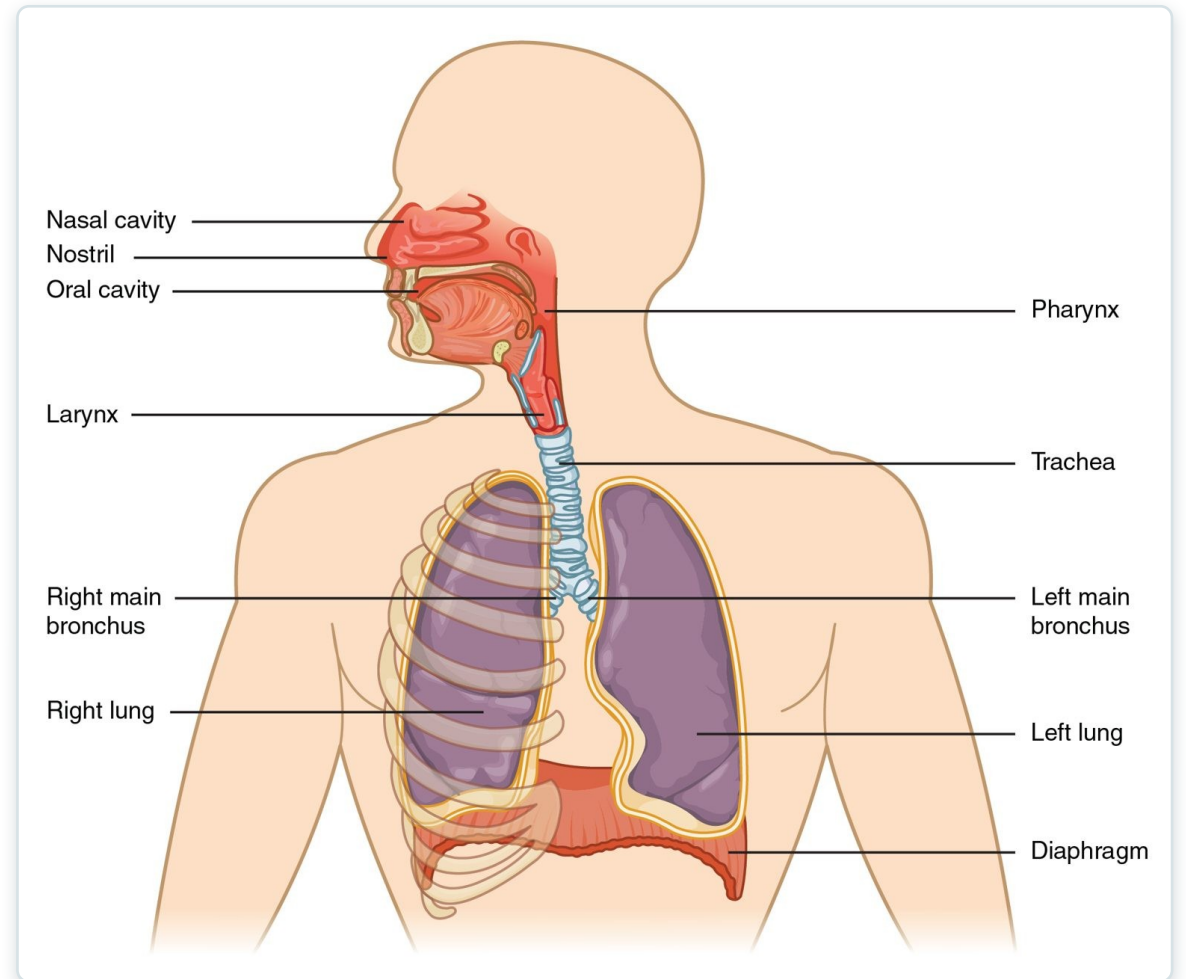
Organs and Structures of the Respiratory System

The airways that carry air from the nose to the alveoli.

SECTION 22.1

The Respiratory System

- Upper and lower respiratory tracts
- Nose, pharynx, larynx, trachea
- Bronchi branch into the lungs
- Ends at the alveoli

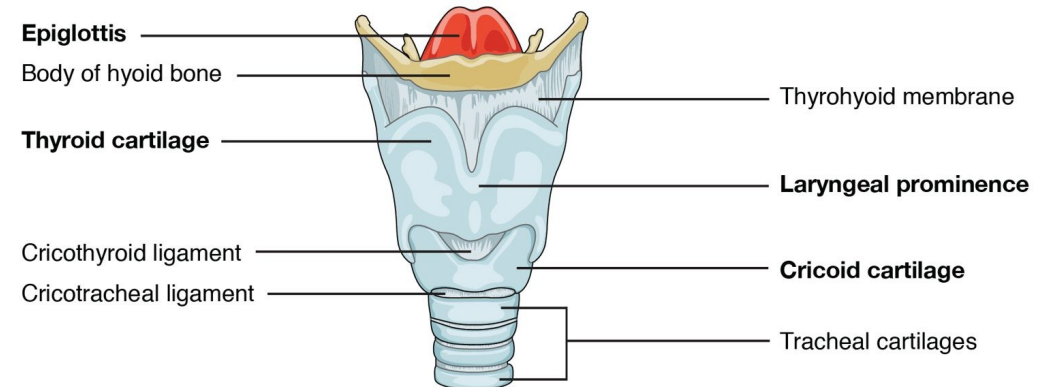


The respiratory system

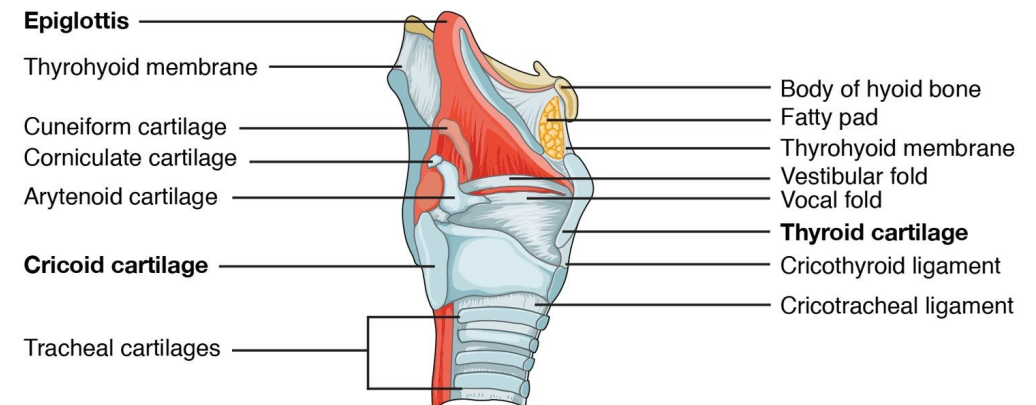
SECTION 22.1

The Larynx

- The "voice box"
- Connects pharynx to trachea
- Houses the vocal cords
- Epiglottis guards the airway



Anterior view

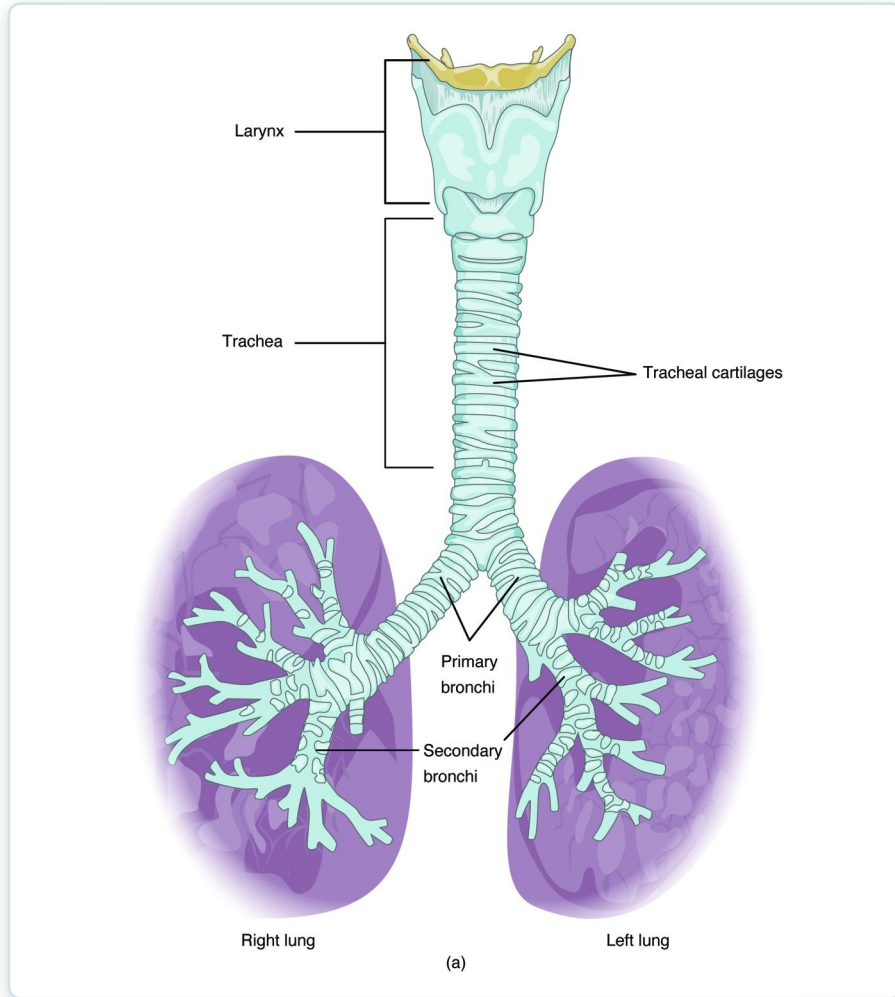


Right lateral view

The larynx

SECTION 22.1

Trachea & Bronchial Tree



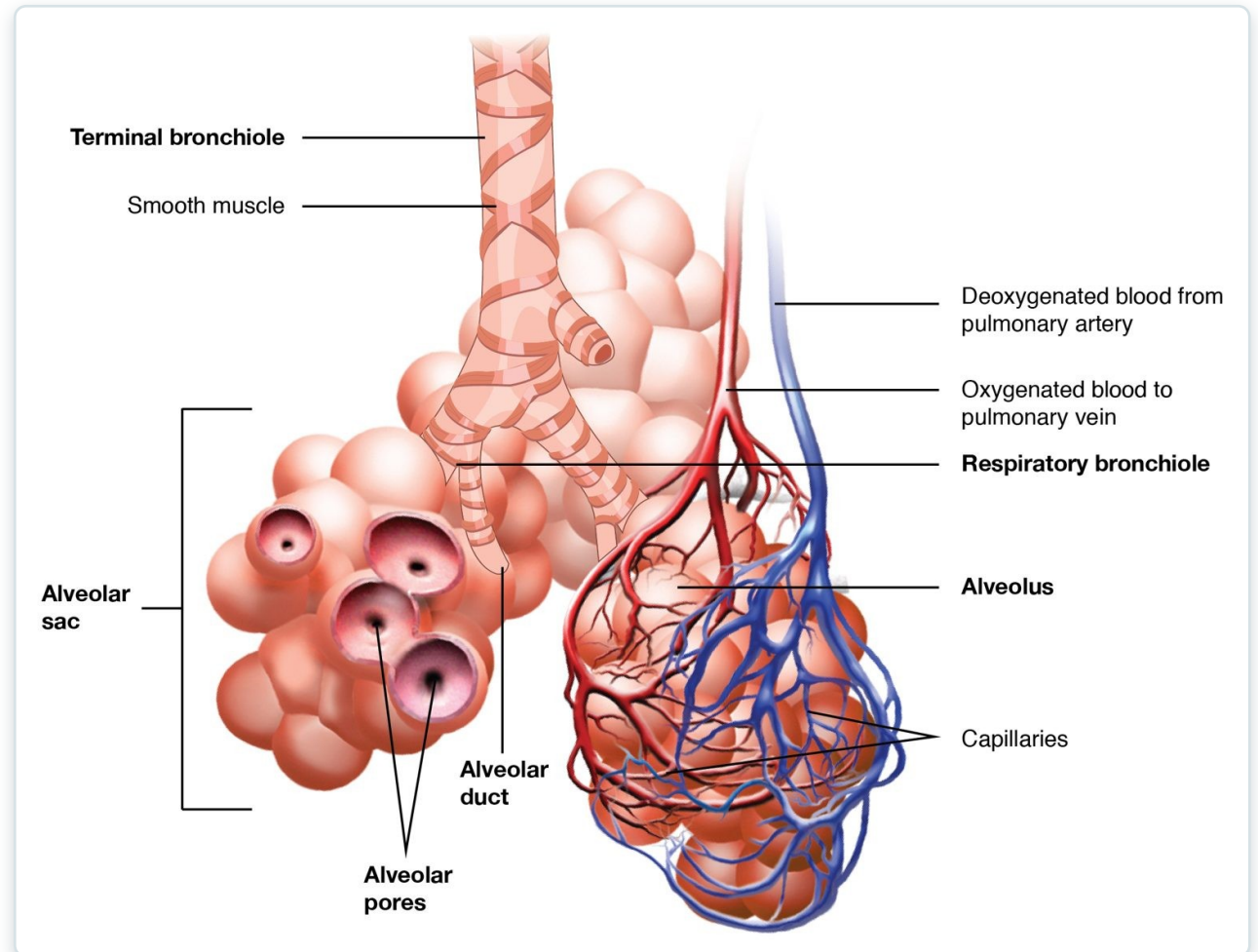
The trachea and bronchial tree

- The trachea is the windpipe
- C-shaped cartilage rings hold it open
- It splits into two main bronchi
- Bronchi branch like an upside-down tree

SECTION 22.1

Bronchioles & Alveoli

- Bronchioles end in alveolar sacs
- Alveoli are tiny air sacs
- Wrapped in capillaries
- Where gas exchange occurs



Bronchioles and alveoli

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22.2

SECTION

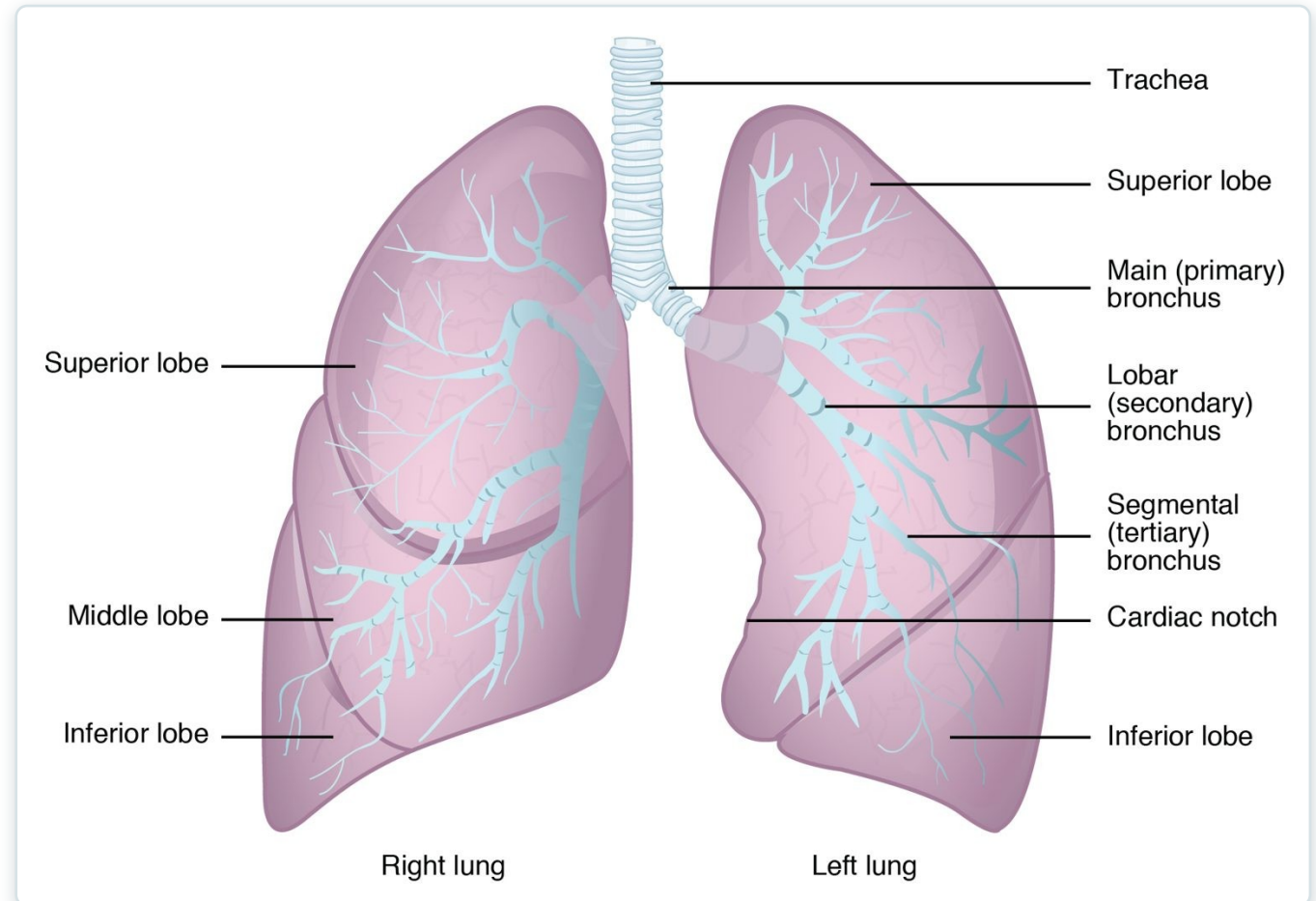
The Lungs

The lungs, their lobes, and the pleura that surround them.

SECTION 22.2

The Lungs

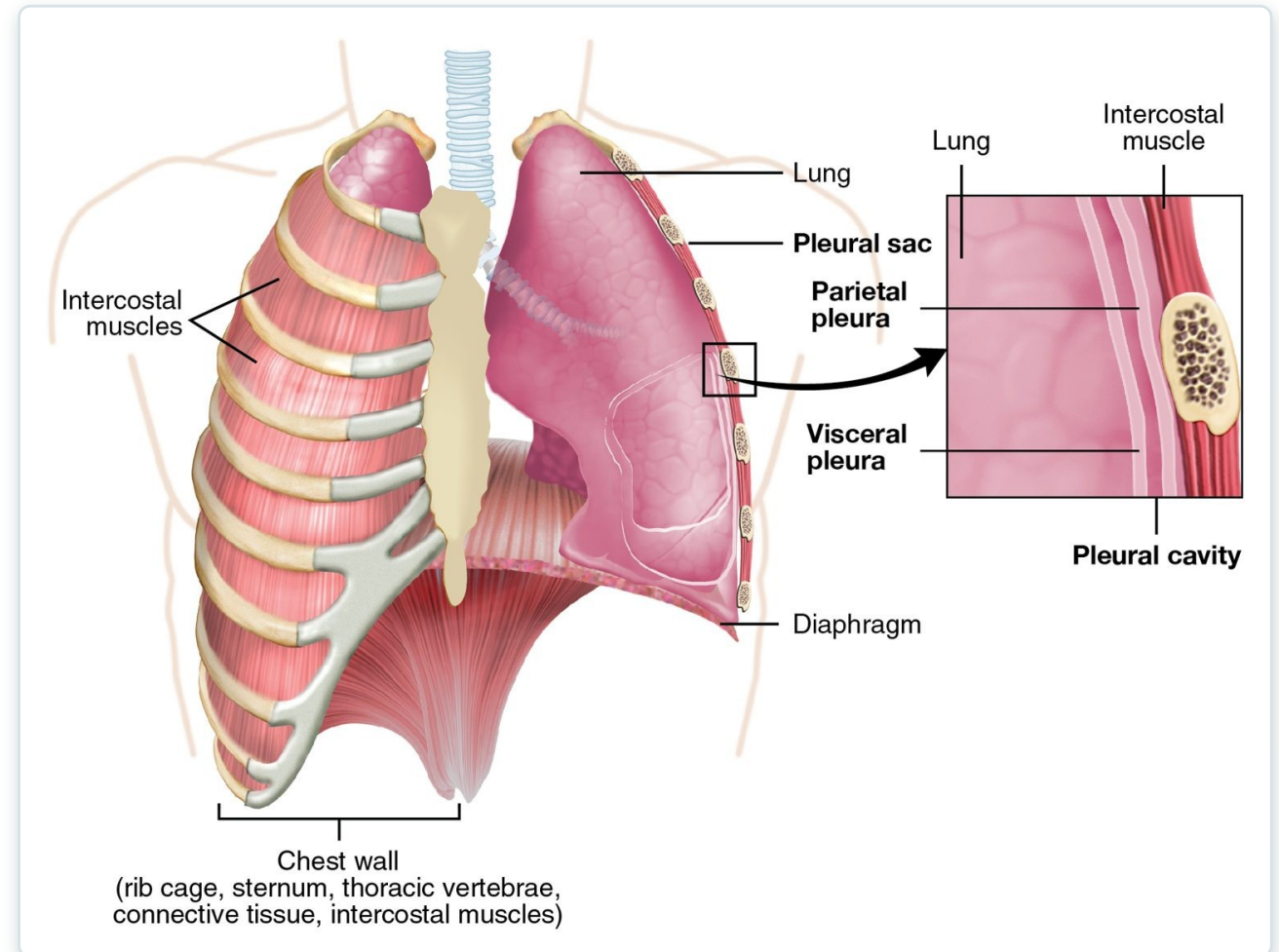
- Two lungs fill the chest
- Right lung has three lobes
- Left lung has two lobes
- Left is smaller (room for the heart)



The lungs and their lobes

The Pleura

- A double membrane around each lung
- Pleural fluid lies between the layers
- Reduces friction during breathing
- Helps the lungs expand



The pleural membranes

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22.3

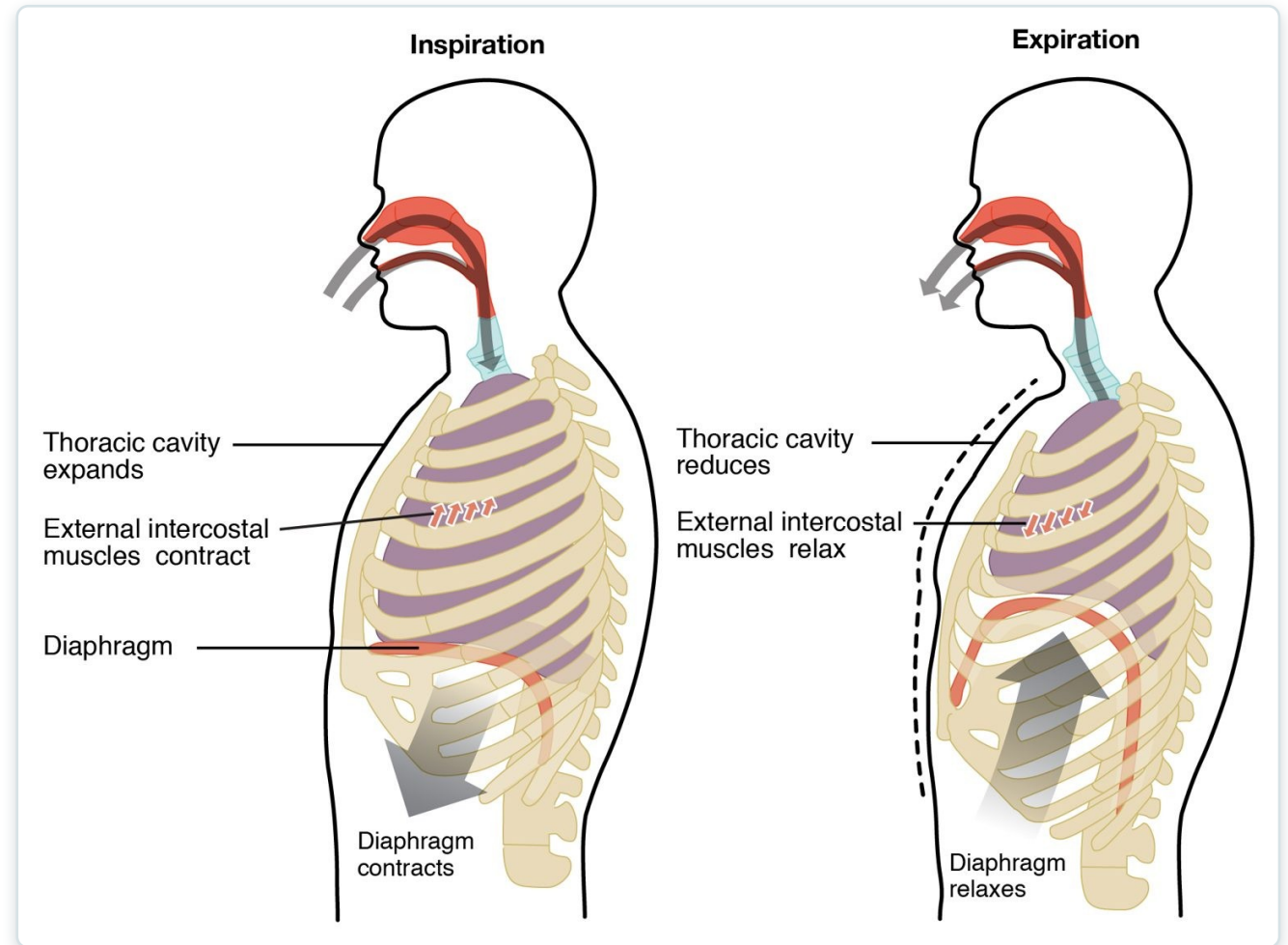
SECTION

The Process of Breathing

How the body moves air in and out of the lungs.

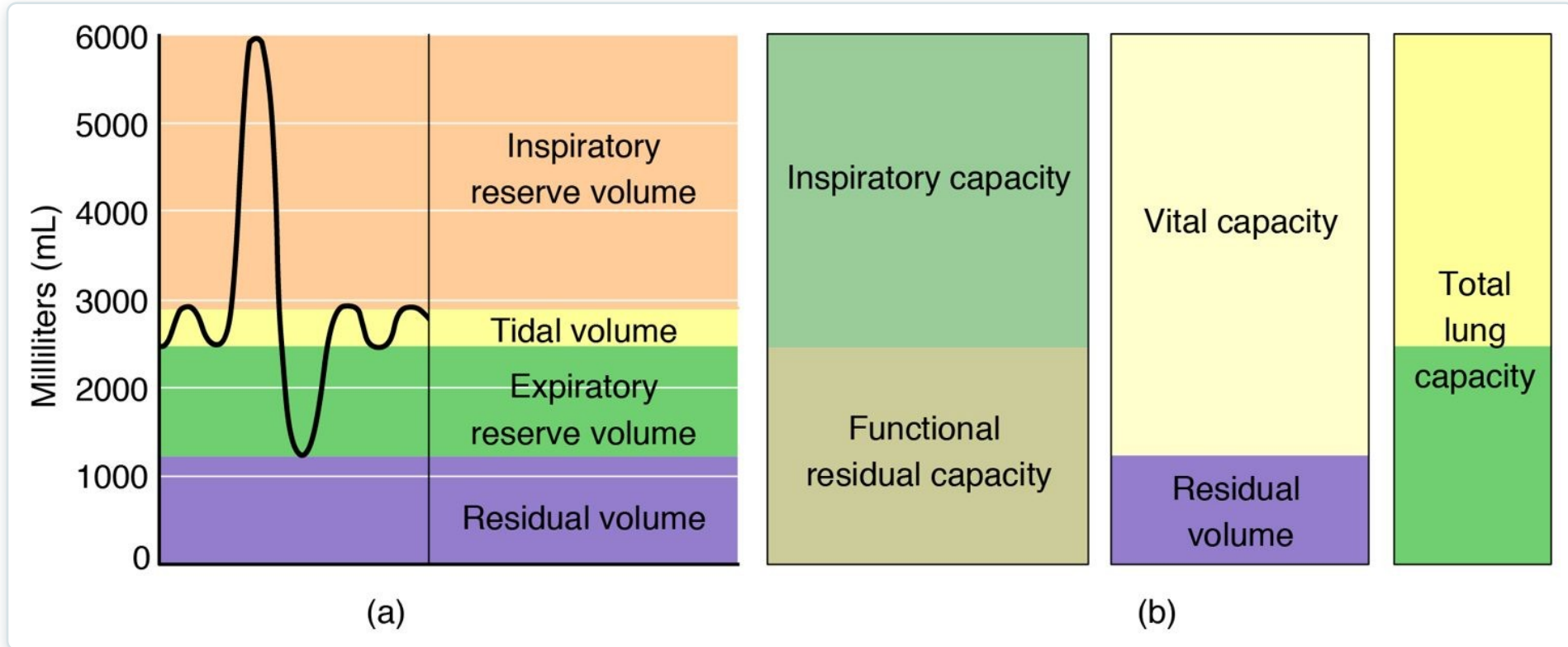
Inspiration & Expiration

- Breathing changes chest volume
- Diaphragm and intercostals contract
- Volume up → pressure down → air in
- Relaxation pushes air out



The mechanics of breathing

Lung Volumes & Capacities

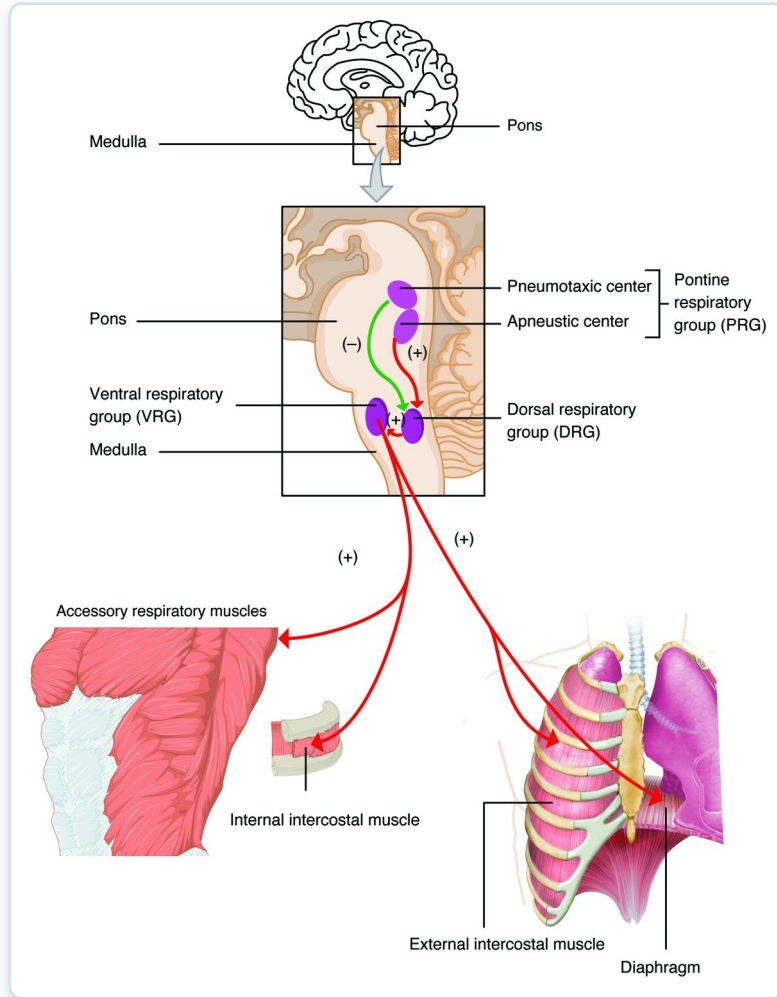


Lung volumes and capacities

- The lungs hold measurable volumes
- Tidal volume = a normal breath
- Vital capacity = max breath out
- Measured by spirometry

SECTION 22.3

Control of Breathing



Neural control of breathing

- The brainstem sets the rhythm
- Centers in the medulla and pons
- Adjust rate and depth automatically
- Respond to CO₂ and oxygen levels



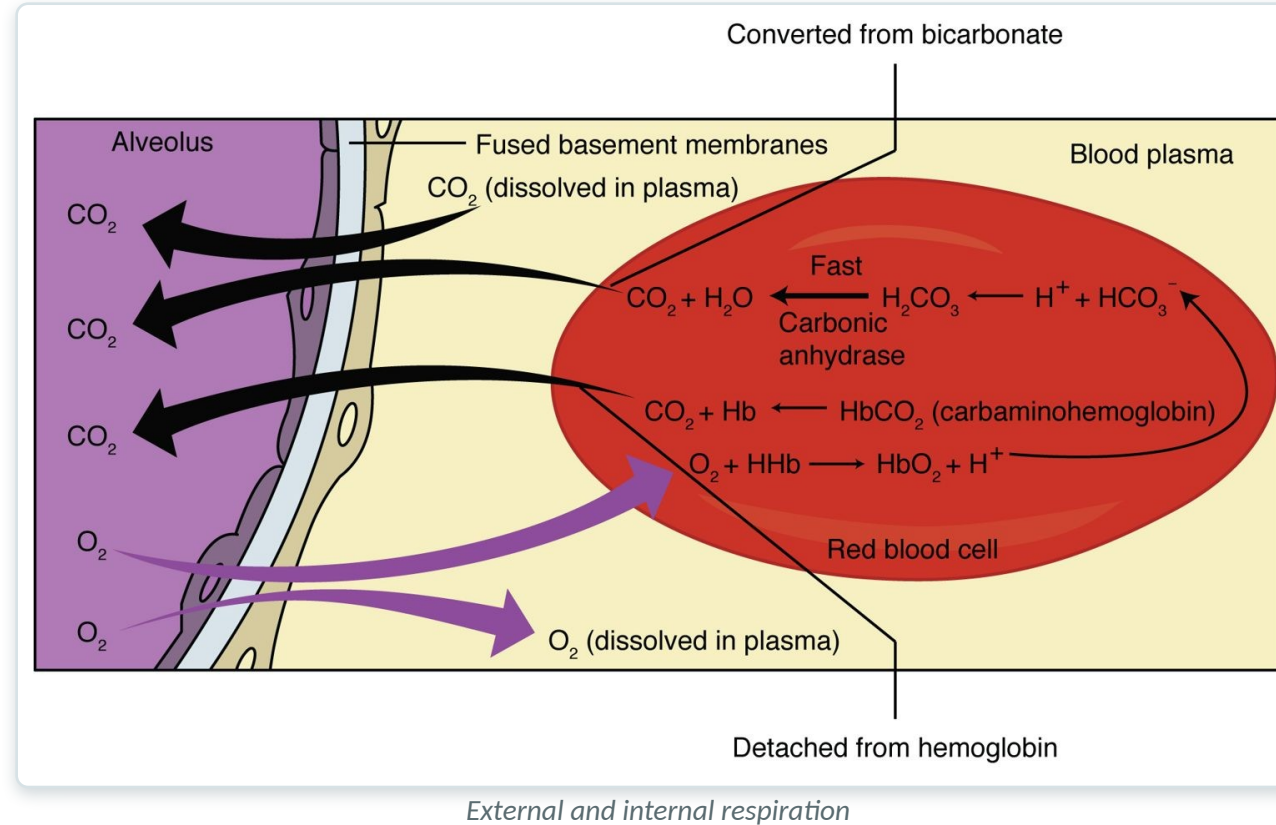
22.4

SECTION

Gas Exchange

How oxygen and carbon dioxide are exchanged in the body.

Gas Exchange



- Gases move by diffusion
- External respiration in the lungs
- O₂ in, CO₂ out at the alveoli
- Internal respiration at the tissues



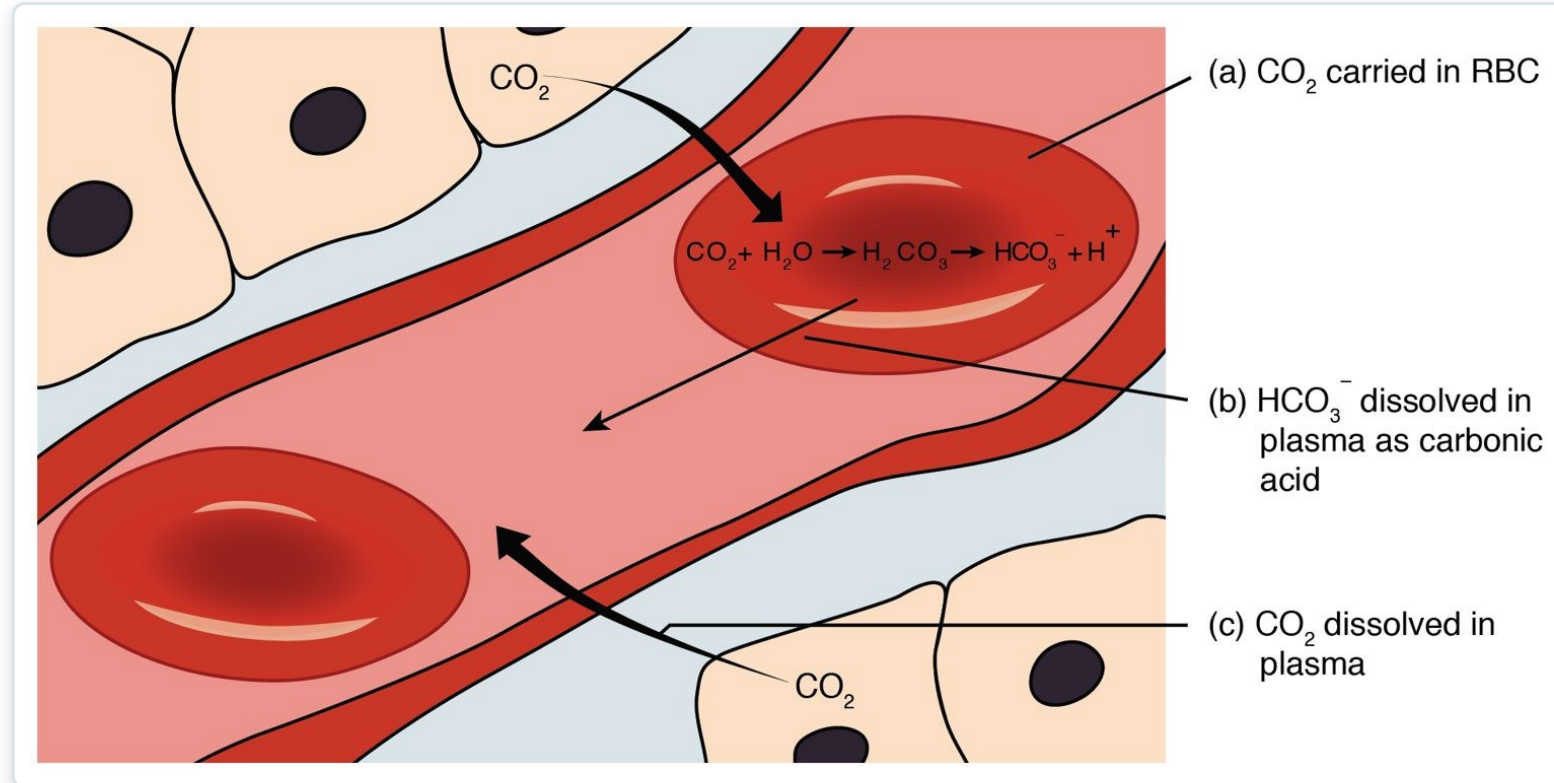
22.5

SECTION

Transport of Gases

How the blood carries oxygen and carbon dioxide.

Transport of Gases



Transport of oxygen and carbon dioxide

- Most oxygen rides on hemoglobin
- Most CO₂ travels as bicarbonate
- Some gas dissolves in plasma
- Carried between lungs and tissues



22.7

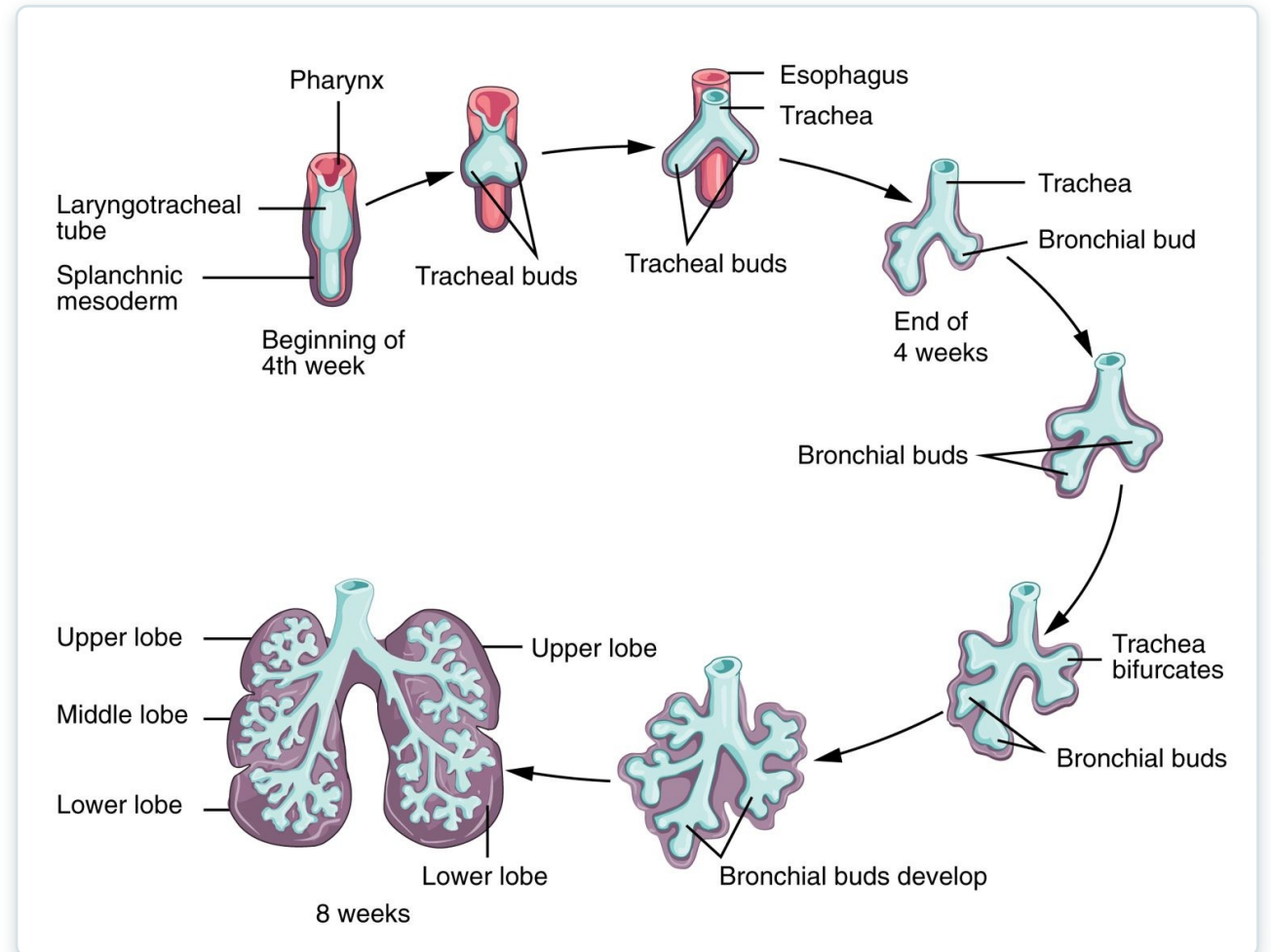
SECTION

Embryonic Development of the Respiratory System

How the lungs and airways form before birth.

Respiratory Development

- The airways begin as a lung bud
- They branch into the bronchial tree
- Alveoli mature late in pregnancy
- Surfactant readies the lungs for air



Development of the respiratory system

Key Takeaways

- The respiratory tract carries air from the nose to the alveoli, where gas exchange occurs.
- The lungs are wrapped in pleura that allow smooth, friction-free breathing.
- Breathing moves air by changing chest volume and therefore pressure.
- Gases are exchanged by diffusion at the lungs and at the tissues.
- Oxygen is carried on hemoglobin and carbon dioxide mostly as bicarbonate.

CLINICAL CONNECTIONS

Why the Respiratory System Matters in Practical Nursing

Asthma & COPD

Narrowed airways make breathing difficult.

Pneumonia

Infection fills alveoli, impairing gas exchange.

Oxygen Therapy

Supplemental oxygen supports patients with low blood oxygen.

Aspiration

Food or fluid entering the airway is a serious risk.

Pulse Oximetry

Measures blood oxygen saturation noninvasively.

Respiratory Rate

A vital sign reflecting breathing and overall status.